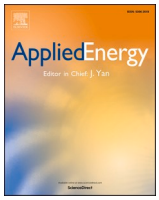
Applied Energy 304 (2021) 117841



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy





# A short- and long-term prognostic associating with remaining useful life estimation for proton exchange membrane fuel cell

Zhendong Zhang [a], Ya-Xiong Wang [a,b,*], Hongwen He [a], Fengchun Sun [a]

[a] National Engineering Laboratory for Electric Vehicles, Beijing Institute of Technology, Beijing 100081, China
[b] School of Mechanical Engineering and Automation, Fuzhou University, Fuzhou 350108, China

## HIGHLIGHTS

- Five multi-step strategies are compared to implement short-term voltage prognostic.
- Variable-step procedures are proposed to forecast the long-term degradation trend.
- Based on the evolutionary algorithm, the hyper-parameters adjustment is proposed.
- Remaining useful life is evaluated under the data-driven prognostics framework.

## ARTICLE INFO

*Keywords:*
Proton exchange membrane fuel cell
Neural network
long short-term memory (LSTM)
Multi-step ahead prognostics
Data-driven model
remaining useful lifetime (RUL)

## ABSTRACT

Proton exchange membrane fuel cell (PEMFC), as a promising power source, provides a feasible solution for clean and low-carbon energy systems. The durability problem restricts PEMFC application in some scenarios, which can be improved by the prognostic technology indirectly. This paper aims to develop a data-based method to implement the short-term and long-term prognostic simultaneously, and the developed long-term prognostic can be performed without future operation information. First, the short-term prognostics of five multi-step ahead forecasting strategies are proposed and compared based on a long short-term memory (LSTM) network. Results show that the multi-step input and multi-step output (MIMO) with LSTM strategy has a better performance in the short-term prognostics under the test conditions of the stationary and dynamic current. Then, the hyper-parameters of the prediction model are determined by an evolutionary algorithm. Furthermore, in the long-term prognostics regime, the variable-step long-term method is proposed and rectified by the short-term prognostics. Finally, the developed remaining useful life (RUL) prediction is compared with a model-based extended Kalman filter. The average root mean square error results for the short-term prognostics of two conditions are 0.00532 and 0.00538, respectively. The RUL estimations of two PEMFCs named FC1 and FC2 are given with 95% and 90% confidence intervals, respectively. Consequently, the proposed method can achieve acceptable accuracies in the short-term prognostic, the long-term prognostic, and the RUL prediction.

## 1. Introduction

As an environmentally friendly power system with high efficiency, the proton exchange membrane fuel cell (PEMFC) has been treated as one of the most promising power sources that is gradually applied in the automobile, stationary product, and portable power areas [1]. However, the material degradation in each component of the PEMFC may result in the shortage of its life span, which constrains the deployment and commercialization. Prognostic is the core technique in the prognostics and health management (PHM) process, which helps to monitor the state of health and estimate the remaining useful life (RUL) of PEMFC [2]. Consequently, even if material degradation is inevitable, an accurate prognostic can provide a foundation of decision-making and control to decelerate the aging process. From different perspectives, prognostics can be divided into short- and long-term prognostic. The short one can catch the local variation with high accuracy, while the long-term prognostic can capture the aging trend. Here, the PEMFC output voltage as a readily available parameter is chosen to evaluate the degree of aging. And the prognostics are conducted without prior knowledge of the future information. Generally, the prognostics of PEMFCs can be distinguished into three approaches: data-based methods, model-based

* Corresponding author at: National Engineering Laboratory for Electric Vehicles, Beijing Institute of Technology, Beijing 100081, China.
*E-mail addresses:* yxwang@bit.edu.cn, yxwang@fzu.edu.cn (Y.-X. Wang).

https://doi.org/10.1016/j.apenergy.2021.117841
Received 30 April 2021; Received in revised form 24 August 2021; Accepted 3 September 2021
Available online 25 September 2021
0306-2619/© 2021 Elsevier Ltd. All rights reserved.

**Nomenclature**

| | |
|---|---|
| $C_t$ | cell state |
| $C_t$ | cell candidate state |
| $H_t$ | cell hidden state |
| $X_t$ | cell input |
| $f_t, i_t, o_t$ | outputs of the each gate |
| $\sigma$ | sigmoid activation function |
| $W_f, W_i, W_o$ | weight of the each gate |
| $b_f, b_i, b_o$ | bias of the each gate |
| $y_t$ | measured voltage |
| $\hat{y}_t$ | predicted voltage |
| $N$ | historical horizon length |
| $H$ | prognostics horizon length |
| $f$ | recursive strategy model |
| $g_i$ | direct strategy model |
| G | MIMO strategy model |
| $F_i$ | DirRec strategy model |
| $\hat{G}_i$ | DIRMO strategy model |
| $V_{st}$ | stack voltage (V) |
| $n$ | number of fuel cells |
| $E_0$ | open-circuit voltage (V) |
| $R$ | total resistance ($\Omega \cdot cm^2$) |
| $a$ | the Tafel constant |
| $b$ | the concentration constant |
| $T$ | stack temperature (K) |
| $i$ | stack current (A) |
| $i_0$ | exchange current ($A/cm^2$) |
| $i_L$ | limiting current ($A/cm^2$) |
| $\alpha(t)$ | degradation parameter |
| $\beta(t)$ | degradation rate |
| PEMFC | proton exchange membrane fuel cell |
| LSTM | long short-term memory |
| RUL | remaining useful life |
| EKF | extended Kalman filter |
| RMSE | root mean square error |
| MAE | mean absolute error |
| FC | fuel cell |
| MIMO | multi-step input and multi-step output |
| EA | evolutionary algorithm |

methods, and hybrid methods [3].

### 1.1. Literature review

Machine learning technique, as one of the data-based methods, is introduced as the black-box model to describe the aging process of the PEMFC. An estimator of solid oxide fuel cells based on a neural network has been developed to predict voltage variation due to the degradation [4]. In multi-scale prediction, Silva et al. [5] proposed a methodology based on adaptive neuro-fuzzy inference systems to achieve short, medium, or long-term predictions. However, this method represents a relatively high error in short-term prediction. Echo state networks (ESN) [6,7] were introduced into the prognostics of the PEMFC together with a filtering method and multi-input parameters, respectively. The prognostics method based on the relevance vector machine (RVM) was proposed in [8] to improve the accuracy in short-term forecasting, which has been validated in the single-step-ahead and multi-step-ahead prediction. Furthermore, a modified RVM was then proposed to meet the condition of relatively small training data sets [9]. Besides, pattern recognition [10] has exhibited the effectiveness of the PEMFC prognostic. Zhou et al. [11] proposed a hierarchical degradation prognostic approach to guarantee the robustness and prediction accuracy. Liu et al. [12] proposed a short-term prognostics method based on the group method of data handling and the wavelet analysis to achieve high accuracy in different load conditions. An adaptive neuro-fuzzy inference system with a fuzzy c-means strategy was applied for the same purpose in [13], which was training with the raw data. Besides, this approach combing with the automatic parameter adjustment has been verified on an experimental platform. An ensemble ESN was implemented to enhance the prognostic adaptability [14]. Based on the virtual steady-state stack voltage health indicator, an acceptable prognostic could be achieved in the long-term prediction horizon. Chen et al. [15] have proposed a novel degradation prognosis based on the wavelet neural network and Cuckoo search algorithm. This method considering the main operating conditions could be applied to the health state prognostic of PEMFC under different conditions. Xie et al [16] developed a hybrid neural network combining the deep belief network and extreme learning machine with the actual data validation. A digital twin technique has been extended to the energy domain to establish an ensemble RUL prediction system [17]. To improve the data processing ability, the attention mechanism helps to extract the useful feature [18] and a self-optimization regression method is introduced to prognose the degradation trend [19]. To enhance the reliability, He et al. [20] proposed a long-term prediction method based on a least absolute shrinkage and selection operator-echo state network together with a model selection rule. And the final prediction was defined as an average value of the prediction results.

The model-based approach is always used to describe the long-term aging process of PEMFC combining empirical and physical models. Jouin et al. [21] have proposed three empirical models, including linear, logarithmic, and exponential models to approximate the aging process. And the prognostic was conducted under a particle filtering framework to estimate the RUL of the PEMFC. Furthermore, the degradation model has been derived from the polarization curve by introducing two indicators of the state of health [22]. The extended Kalman filter (EKF) has helped to forecast the future performance and RUL of the fuel cell. More health indicators have been proposed. Zhou et al. [23] presented a multi-physical aging model, considering ohmic losses, reaction activity losses, and mass transfer losses. A prognostics framework composed of a parameter extraction process and the particle filtering method was proposed which realized the RUL prediction at a constant operating condition [24]. Zhu et al. [25] applied the Gaussian process state space to infer the internal time-varying parameters of the dynamic model. In addition, an observer-based prognostic algorithm based on the EKF was applied to estimate the actual state of health under variable loads and verify the reliability by quantifying the uncertainty [26]. From the other perspective, Jha et al. [27] developed a special model of PEMFC by using Bond Graph theory, which presented a simplified graphical representation of the complex physics of PEMFC and efficient decomposition based on the functionality. The prognostics problem can be formulated as the joint state-parameter estimation problem in the particle filter framework where the estimation is obtained in the probabilistic domain as well. However, the stack voltage is only related to degradation symptoms and is significantly affected by operating conditions. To improve the prognostics accuracy, the approach coupling of two numerical models concerning the fuel cell operation and surface degradation has been investigated in [28]. Chen et al. [29] selected the stack voltage and power to construct a fusion health indicator. Then, a 2nd order Gaussian degradation model was built to complete the RUL prognostics based on an unscented particle filter. Two different physics-based models were used for the objective of combining the stack voltage and the stack state of health as a fusion health indicator, and an

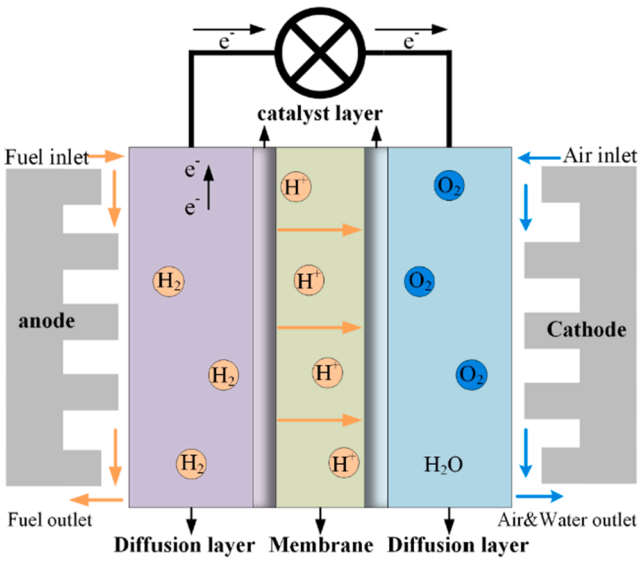



**Fig. 1.** The single-cell elementary structure of PEMFC.

ensemble approach was applied to further implement the aggregation of the RUL prediction and RUL uncertainty under the particle filtering framework [30]. According to a multi-scale modeling approach, Robin et al. [31] introduced a mechanistic catalyst dissolution model coupled with a dynamic PEMFC cell model to predict the equivalent loss of active surface area loss using an innovative numerical method. As for automobile application, a semi-empirical model-based prognostics method could achieve degradation prediction in [32]. Electrochemical surface area, equivalent resistance, and a recovery factor have been introduced into the degradation model. Chen et al. [33] developed an unscented Kalman filter algorithm for degradation prediction with both high accuracy and low computational cost, and it is the first time that field-based data was used.

The hybrid approaches are the integration of the previous two methods. An improved prognostic model was developed to predict the RUL, which combined a hidden semi-Mark model with an empirical model [34]. Cheng et al. [35] proposed a hybrid prognostic framework combining with least square support vector machine and regularized particle filter. The observation values in the prognostic process were obtained by the data-based method. Moreover, a brand-new perspective of the hybrid degradation model was raised in [36]. The voltage aging waveform of PEMFC was decomposed into multiple sub-waveforms by the wavelet analysis. Then, each sub-waveform could be characterized by the extreme learning machine. The moving window technique was used to adjust the fusion weight factors of model-based and data-based approaches adaptively. This method could capture the long-term degradation trend, and the local voltage variation characteristics as well [37]. Xie et al. [38] proposed a fusion prognostic approach based on particle filter and LSTM neural network which could be more robust to the external disturbance.

### 1.2. Motivation and innovation

Back to the data-based method, some of the prognostics methods are just focusing on the short-term forecasting field by introducing adaptive techniques [8–9,12–13,39]. In [40], the LSTM network was adopted to achieve high accuracy in single-step ahead prediction, while the RUL was defined as the difference value between the time to predict and the End-of-Life (EOL) point of the prediction curve [15]. In the aspect of long-term forecasting, the RUL prediction is achieved with iteration predictions generally. On the one hand, this recursive strategy may lead to an accumulation of errors by introducing the predicted voltage as the training input. On the other hand, the use of the operation parameters (such as current, inlet, and outlet temperature) as inputs of the prognostic model is a hypothesis that states that the future information is known as prior knowledge, which is not always available in reality [20,41–42].

In this paper, the voltage time-series datasets are used to describe the aging process of PEMFCs. The LSTM network is used to fulfill short-term prognostic and life prediction. Secondly, the hyper-parameters adjustment method based on the evolutionary algorithm (EA) is used. Furthermore, the accuracy of different multi-step ahead short-term prognostics strategies of PEMFCs are compared, and to find the most suitable strategy. Moreover, the proposed Multi-step Input and Multi-step Output (MIMO) strategy is extended into the long-term prognostics field and implement RUL prediction at the same time.

(1) Based on the LSTM network, the short-term prognostics of five multi-step ahead forecasting strategies are discussed under two PEMFC experiment datasets with the stationary and dynamic current.
(2) A hyper-parameters adjustment method based on the EA is deployed to improve the forecasting performance, including the minimum batch size, the training epoch, and the cell numbers of the hidden layers.
(3) The degradation trends of two stacks are characterized by the variable-step long-term prognostics method with the MIMO strategy. Besides, the RUL prediction is achieved under this prognostics framework.

### 1.3. Organization

The core of the paper is organized as follows. A description of the raw data source, the experimental setup, and the performed test can be found in Section 2. In section 3, the prediction method based on the LSTM is briefly outlined at first. Then, five multi-step short-term forecasting strategies of PEMFC are introduced. At last, a variable-step long-term forecasting method is extended from the MIMO strategy. In section 4, the results of different short-term prognostic strategies, the variable-step long-term prognostic, and RUL prediction are discussed. Finally, conclusions are presented.

## 2. Experiment and dataset analysis

In this section, a brief introduction of PEMFC operation principles is given at first. Then, the key information of the aging experiment settings and the aging dataset preprocessing is presented.

### 2.1. Fundamentals of the PEMFC

As an electrochemical device, the principle and the composition of the fuel cell are shown in Fig. 1 using oxygen and hydrogen to produce electricity, water, and heat. The global reaction equation of the system is:

$$2H_2 + O_2 \rightarrow 2H_2O + electricity + heat \quad (1)$$

Due to the limitation on the capacity and the rated power, cells are generally integrated into a stack in series to cover the electricity demand. As a power source, the stack should be equipped with different auxiliaries to implement gas supplement, control, and power output.

### 2.2. Experiment configuration and dataset analysis

The PEMFC aging experiment is implemented by the Federation for Fuel Cell Research (FCLAB), which contains two aging datasets including Electrochemical Impedance Spectroscopy (EIS) data, and polarization data [43]. Physical parameters involved in the stack can be measured and controlled by the homemade interface in the LabVIEW® and the National Instrument Control system, respectively [43]. The

**Table 1**
Operating parameters.

| Parameter | Value |
|---|---|
| Number of cells | 5 |
| Relative humidity of the anode and the cathode | 50% |
| Temperature | 55 ℃ |
| Stack current | 70 A (FC1) |
| | 70 A with 7 A oscillations (FC2) |
| Relative humidity | 50% (FC1) / 52% (FC2) |
| The gas pressure | 1.3 bar |

PEMFC is realized with commercial membranes, diffusion layers, and machined flow distribution plates, which are composed of 5 cells with an area of 100 $cm^2$, and the fuel cell operated in stationary regime (FC1) has a constant current of 70 A, and the fuel cell operated under dynamic current load (FC2) has a nominal current of 70 A with 7 A oscillations and a 5 kHz frequency [43]. The two aging experiments of PEMFC are conducted under different operating parameters [43], as summarized in Table 1. Besides, the voltage recovery phenomenon is caused by the polarization curve test and EIS measurement.

Notably, the raw voltage data contains certain noise and sharp peaks, and a large collection of data points may increase the computation time. Therefore, to obtain the degradation trend more effectively, the pre-process is necessary, including outlier handling, data filtering, and datasets splitting. The Gaussian weighted moving average filter is applied to smooth the raw data. The essence of the moving average filter is a kind of low-pass filtering. Its purpose is to filter out high-frequency perturbations and preserve useful low-frequency trends in time series. Afterwards, the two new voltage aging datasets with an interval of 1 h for FC1 and FC2 are obtained as shown in Fig. 2. These two new aging data sequences contain 1154 and 1020 voltage data, which are used to realize the short-term and the variable-step long-term prognostic. In addition, before the prognostics process, it is obvious that the voltage degradation of FC1 test is much more normal and stable than FC2.

## 3. The framework of the prognostics method

The overall framework of the prognostic method put forward in this paper is shown in Fig. 3. The LSTM network is introduced to fulfill the multi-step short-term prognostic first, and the prediction performance of five different strategies is compared under two experiment datasets of PEMFCs from different aging tests. Furthermore, the variable step long-term aging prognostic is achieve based on the MIMO strategy. Both short-term and long-term prognostics are conducted under the EA, which helps to adjust the high parameters of the network instead of the manual regulation.

### 3.1. LSTM network model

Facing the problem of long-term dependencies, LSTMs are specifically designed to avoid this issue as a variant of the recurrent neural network. Thus, remembering information for a long time is their inherent ability, rather than learning from model tuning and training. Due to the chain-like property, LSTM is available in time series prediction, such as the data type of sequences and lists. As a result, LSTM has been used in speech recognition, language modeling, translation, image captioning, and other scenarios.

The cell structure of LSTM is shown in Fig. 4. The key to LSTM is that the previous cell state $C_{t-1}$ to the current cell state $C_t$ can be maintained under the tiny linear interactions. As a result, the cell state preserves valid information over long time scales. To be more specific, the LSTM can remove the old information and add something new, through the structures called gates, including forget gate, input gate, and output gate. The forget gate can be written as:

$$f_t = \sigma\left(W_f \bullet [\mathrm{H_{t-1}}, \mathrm{X_t}] + \mathrm{b_f}\right) \tag{2}$$

where $f_t$ is the forget gate; $\sigma$ is the sigmoid activation function; $W_f$ denotes the weight for forget gate network; $b_f$ denotes the bias of the forget gate network.

In the condition that $f_t$ is 1 and $i_t$ is 0, the cell state copies the content

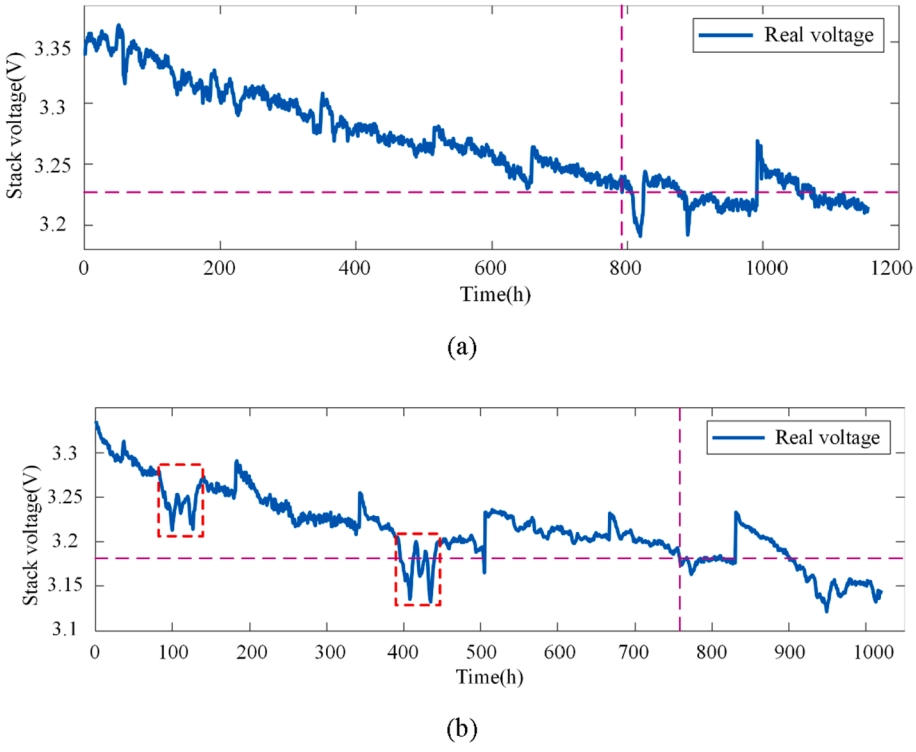



**Fig. 2.** The stack voltage degradation (a) FC1; (b) FC2.

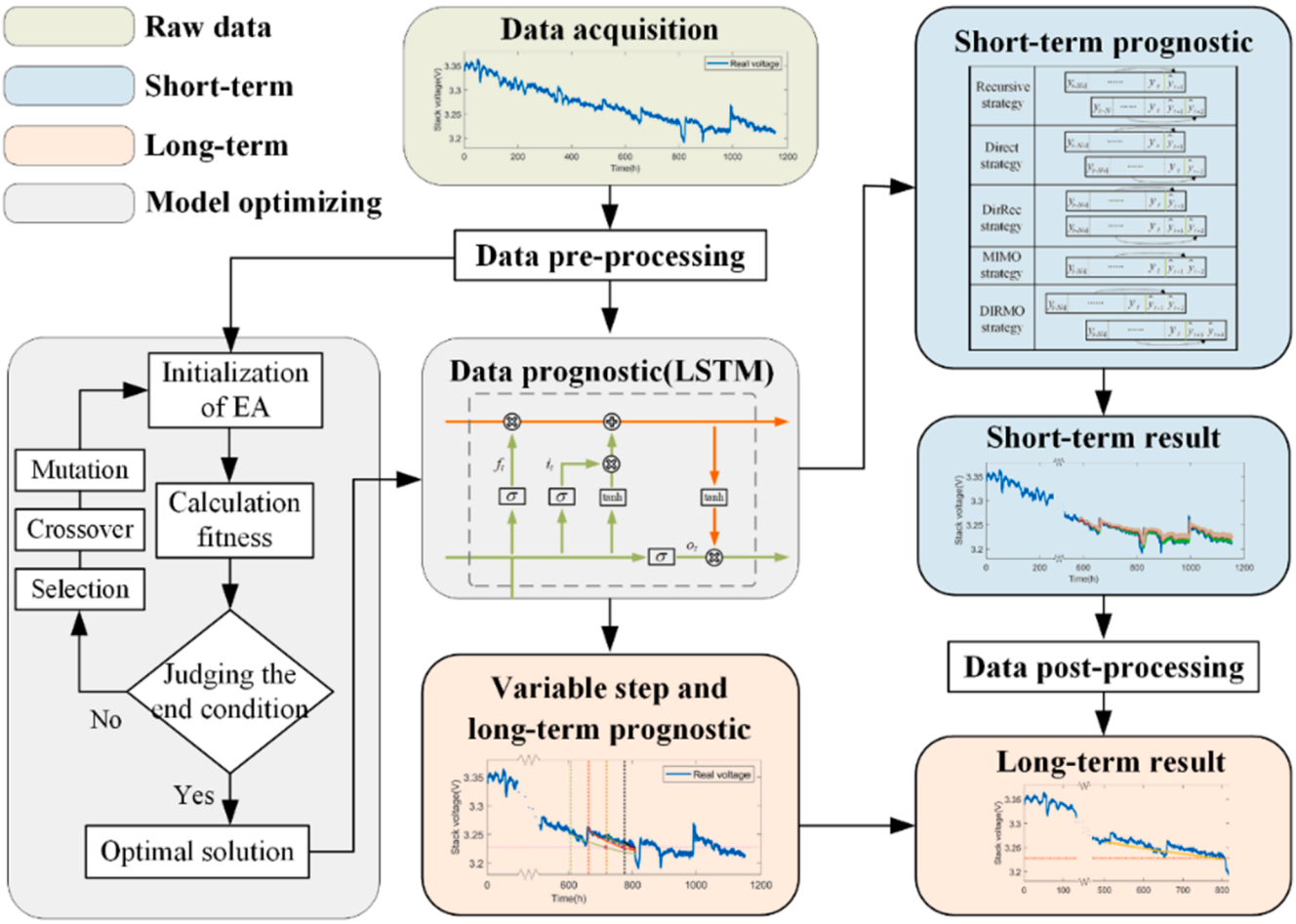



**Fig. 3.** The overall prognostic method scheme.

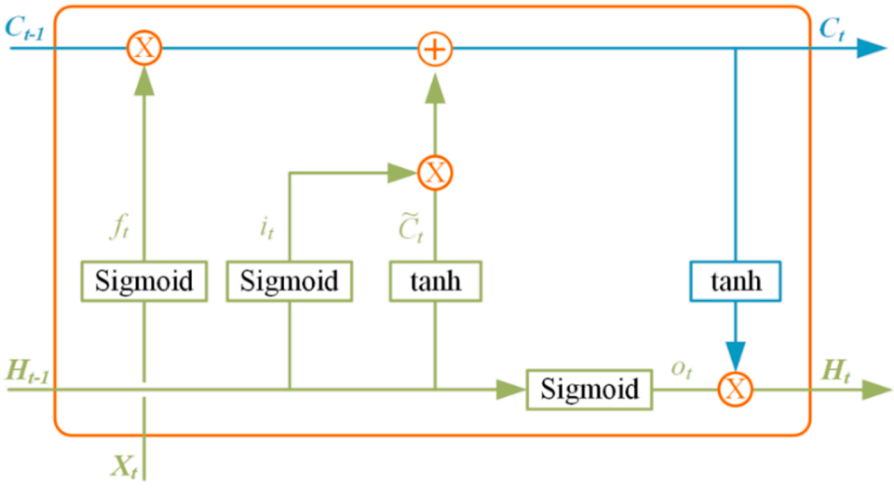



**Fig. 4.** Long Short-Term Memory network cell model.

in the previous step. On the contrary, the candidate state $C_t$ takes place of the history information, which can be described by the following equation.

$$i_t = \sigma(W_i \bullet [\mathrm{H_{t-1}}, \mathrm{X_t}] + \mathrm{b_i}) \tag{3}$$

$$C_t = \tanh(W_C \bullet [\mathrm{H_{t-1}}, \mathrm{X_t}] + \mathrm{b_C}) \tag{4}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{5}$$

where $i_t$ is the input gate; *tanh* is the hyperbolic tangent activation function; $W_i$ and $W_C$ denote the weights of the input gate and the candidate state; $b_i$ and $b_C$ denote the biases of the input gate and the candidate state, respectively; $\odot$ is the element-wise multiplication. Then the output gate $o_t$is given as:

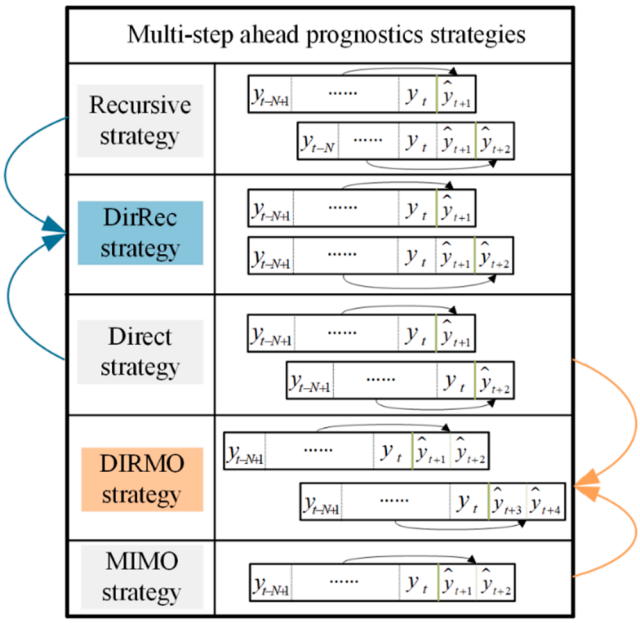



**Fig. 5.** Schematic diagram of the five forecasting strategies in the short-term prognostic.

$$o_t = \sigma(W_o \bullet [\mathrm{H_{t-1}}, \mathrm{X_t}] + \mathrm{b_o}) \tag{6}$$

where $W_o$ denotes the weight of the output gate; $b_o$ is the bias of the output gate; $X_t$ is the input data of the present step. Furthermore, the output $H_t$which is hidden state of the LSTM cell can be written as:

$$H_t = o_t \odot tanh(C_t) \tag{7}$$

### 3.2. Multi-step ahead short-term forecasting strategies

This section will first give a presentation of the five forecasting strategies. Several multi-step ahead forecasting strategies for time series have been proposed and summarized according to relative literature [44], which have been introduced into the multi-step short-term prognostics of the PEMFC system.

(1) Recursive strategy

In this strategy, only one single model $f$ is trained to realize a one-step-ahead prediction.

$$\hat{y}_{t+1} = f(y_t, \cdots, y_{t-N+1}) \tag{8}$$

where $N$ is the historical horizon length; $t$ is the prediction time; $y_t$ is the measured voltage; $\hat{y}_t$ is the predicted voltage. Considering the multi-step ahead forecast, the predicted value is added in the previous step into the same model $f$ for the forecasting of the next step. Thus, this process is executed until every step is forecasted.

(2) Direct strategy

Different from the recursive strategy, the direct strategy forecasts each horizon independently with $H$ models $g_i$ as follows.

$$\hat{y}_{t+i} = g_i(y_t, \cdots, y_{t-N+1}) \tag{9}$$

with $i \in \{1, \cdots, H\}$.

where $H$ is the prognostics horizon length. In this strategy, the accumulation of errors has been avoided by using the real values to conduct the prediction process. However, the assumption of conditional independence can affect the accuracy when the predicted variables remain complex dependencies. Besides, this strategy has suffered from a larger computational burden compared with other strategies, and the complexity is proportional to the prognostics horizon length.

(3) MIMO strategy

The MIMO strategy is introduced to forecast the multi-step voltage with one trained model G at a time and taking the multi-step historical voltage data as an input.

$$\left[\hat{y}_{t+H}, \cdots, \hat{y}_{t+1}\right] = \mathrm{G}(y_t, \cdots, y_{t-N+1}) \tag{10}$$

This strategy avoids the accumulation of errors as well. Different from the Direct strategy, the MIMO strategy preserves the stochastic dependency between two neighboring time steps [44]. However, the flexibility is constrained by using one trained model when conducting the short-term prognostic.

(4) DirRec strategy

The DirRec strategy combines the ideas of the Recursive and the Direct strategies. At each time step, $H$ models $F_i$ are trained to forecast each horizon. Unlike the previous strategies, the historical horizon length is varying from $N$ to $N + H - 1$.

$$F_1 = f \tag{11}$$

$$\hat{y}_{t+i} = F_i\left(\hat{y}_{t+i-1}, y_t, \cdots, y_{t-N+1}\right) \tag{12}$$

with $i \in \{2, \cdots, H\}$. This strategy preserves all the predicted values as input, thus, to a certain extent, achieves the characterization of complex independence. But there is still the phenomenon of error accumulation. In the same way, this strategy suffers from high computational complexity as well.

(5) DIRMO strategy

The DIRMO strategy combines the features of the Direct and MIMO strategies. As a compromise, the multi-step ahead prediction is decomposed into $n$ parts where $n$ models are conducting the MIMO strategy.

$$\left[\hat{y}_{t+pi}, \cdots, \hat{y}_{t+p(i-1)1}\right] = \hat{\mathrm{G}}_i(y_t, \cdots, y_{t-N+1}) \tag{13}$$

with $H = p \bullet \mathrm{n}$ and $i \in \{1, \cdots, n\}$. To increase the flexibility of the prognostic, the stochastic dependency in the forecasting is partly sacrificed by introducing the Direct strategy. And this is the motivation of the DIRMO strategy. However, these strategies have different adaptability to different time-series forecasting problems.

The relationships between different short-term forecasting strategies and the schematic diagram of model iteration are shown in Fig. 5.

### 3.3. Variable-step long-term forecasting method

With the prediction length increasing, the indeterminacy of long-term prognostics will accumulate to reduce the forecasting accuracy and consequently affect the applicability of the method, as shown in Fig. 6. Although, the MIMO strategy can preserve the stochastic dependency between two neighboring time steps.

Due to the small sample training and long-term time-domain prediction as well, the MIMO strategy has the problem of the drift of the initial prediction value. Fortunately, on the account of the accuracy of the short-term forecasting results, the initial values of the long-term forecasting can be corrected by translation as below. The correction process of the long-term forecasting can be described by the following

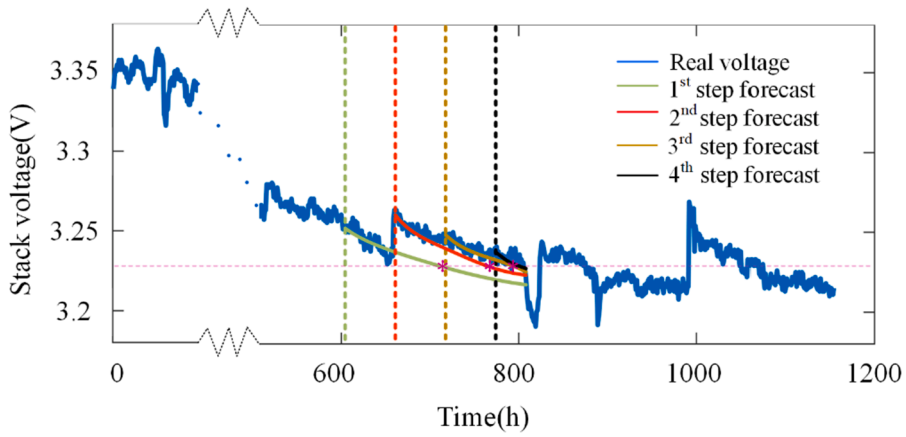



**Fig. 6.** Schematic diagram of the variable-step long-term method.

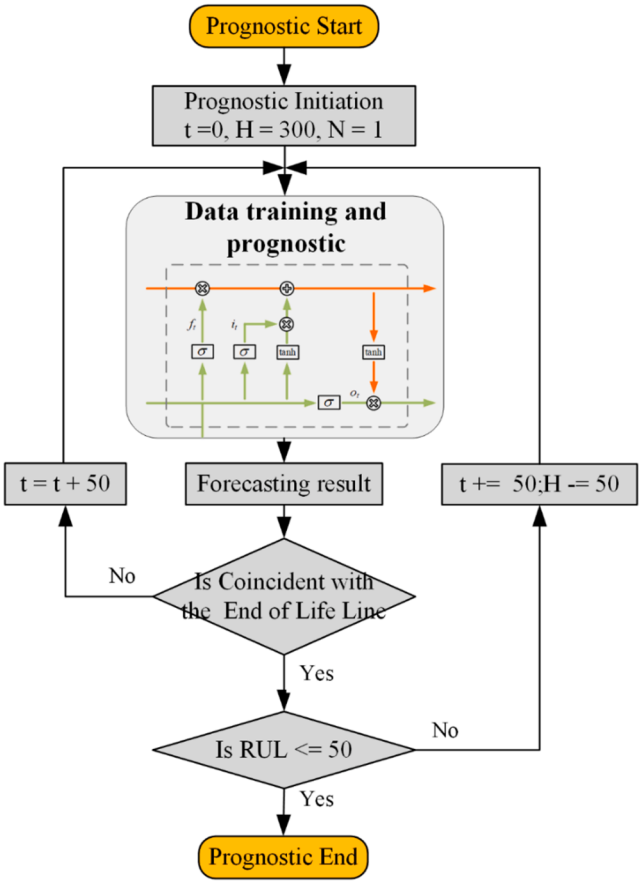


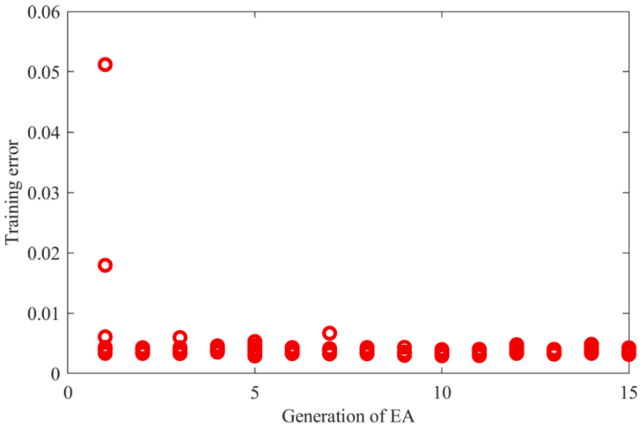

**Fig. 7.** Flow chart of the RUL prediction based on the variable-step long-term method.

**Table 2**
The procedures of the EKF.

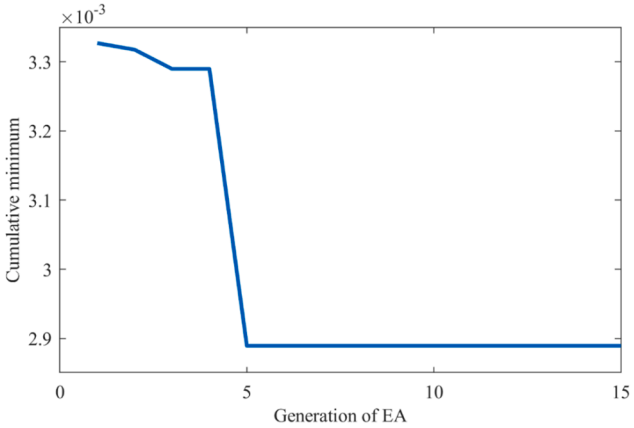
| Algorithm step | Formula |
|---|---|
| (1) Initialization | $X_{0\|0} = E[x_0]$, $P_{0\|0} = E[(X_0 - X_{0\|0})(X_0 - X_{0\|0})^T]$ |
| (2) Prediction | $X_{k\|k-1} = AX_{k-1\|k-1}$, $P_{k\|k-1} = AP_{k-1\|k-1}A^T + Q$ |
| (3) Update | $K_k = P_{k\|k-1}H_k^T(H_kP_{k\|k-1}H_k^T + R)^{-1}$, $H_k = \partial g(X_{k\|k-1}, U_k)/\partial X_{k\|k-1}$ |
| | $P_{k\|k} = (I - K_kH_k)P_{k\|k-1}$, $X_{k\|k} = X_{k\|k-1} + K_k(Z_k - g(X_{k\|k-1}, U_k))$ |

**Table 3**
Parameters of EA.

| Parameter | Value |
|---|---|
| Size of population | 20 |
| Maximum iteration | 15 |
| Probability of mutation | 0.001 |

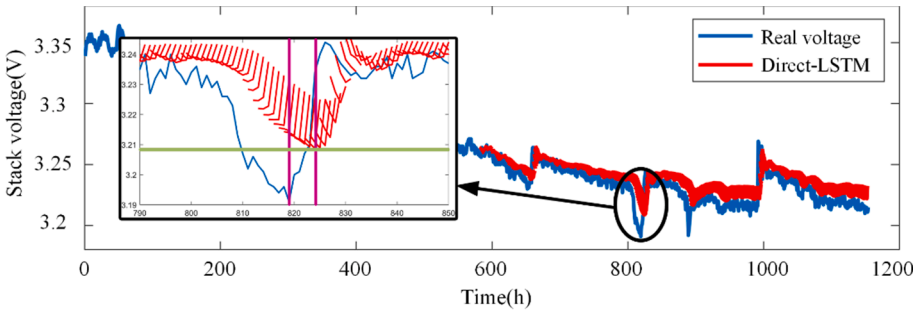



(a)

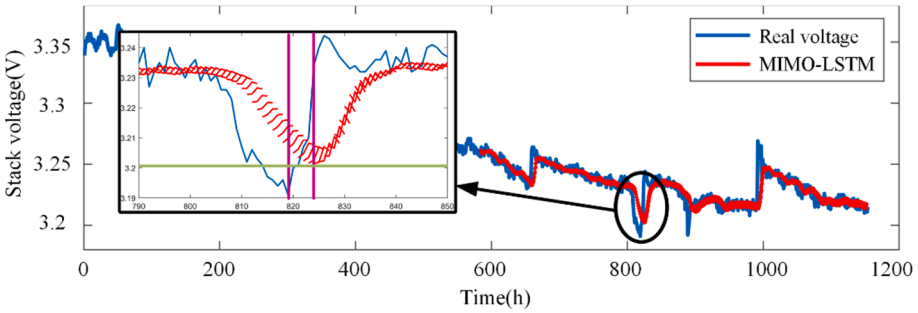



(b)

**Fig. 8.** The structure optimization results of the MIMO-LSTM model: (a) the training errors of the population; (b) the cumulative minimum errors.




(a)




(b)

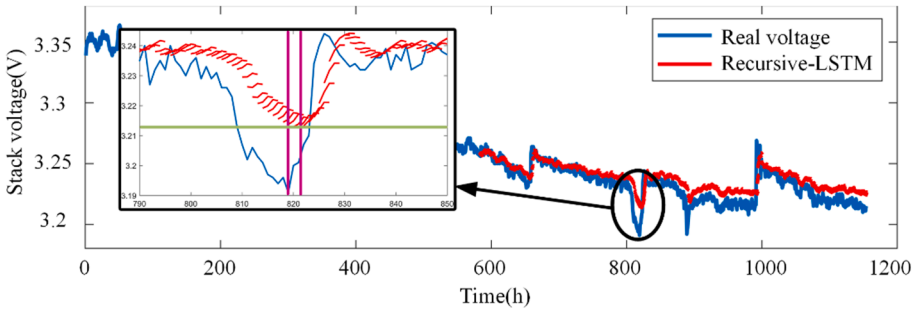



**Fig. 9.** The multi-step short-term forecasting results of FC1: (a)Direct strategy, (b) MIMO strategy, (c) Recursive strategy.

equation.

$$\Delta = \frac{1}{N_y}\sum_{i=1}^{N_y}\left(\widehat{Y}(i) - \widehat{y}(i)\right) \tag{14}$$

$$\widehat{Y}^{-} = \widehat{Y} - \Delta \tag{15}$$

where $N_y$ is the prognostic horizon length of the short-term forecasting; $\widehat{Y}$ and $\widehat{y}$ are the predicted voltage sequences of the long-term and short-term forecasting, respectively; $\Delta$ is the long-term correction factor, and $\widehat{Y}^{-}$ is the predicted voltage sequence after correction. As a result, the life degradation trend obtained by the long-term time-domain prediction can be used to predict the RUL of PEMFC at present.

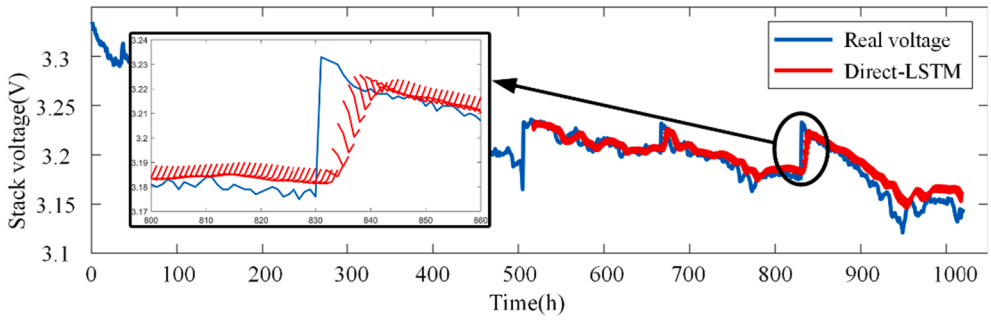
To increase the RUL prediction accuracy of PEMFC, a variable-step long-term forecasting method is proposed. In the beginning, the long-term prognostics length $H$ is set and fixed. While the long-term forecasting sequence is overlapping with the EoL, the prognostics length $H$ will decrease in the next prognostic step to adapt to the RUL prediction in a shorter time domain, as shown in Fig. 7. Here, for example, the forecasting interval is set and fixed as 50 h. In the iteration process, the training datasets are enlarged as well. Besides, the prognostics length is reduced to weaken the uncertainty of the RUL prediction to some extent. The prognostics process ends when the predicted RUL falls below the lower limit. As a result, the variable-step long-term forecasting method can improve the accuracy of the RUL prediction.

### 3.4. Model-based prognostic

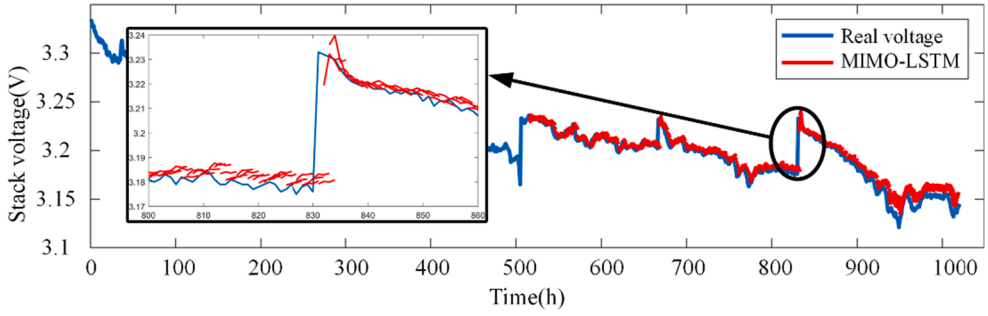
The model-based EKF method is applied to compare with the RUL prediction performance of the variable-step long-term method. The semi-empirical degradation model is derived from the polarization equation as below.

$$V_{st} = n\left(E_0 - Ri - aTln\left(\frac{i}{i_0}\right) - bTln\left(1 - \frac{i}{i_L}\right)\right) \tag{16}$$

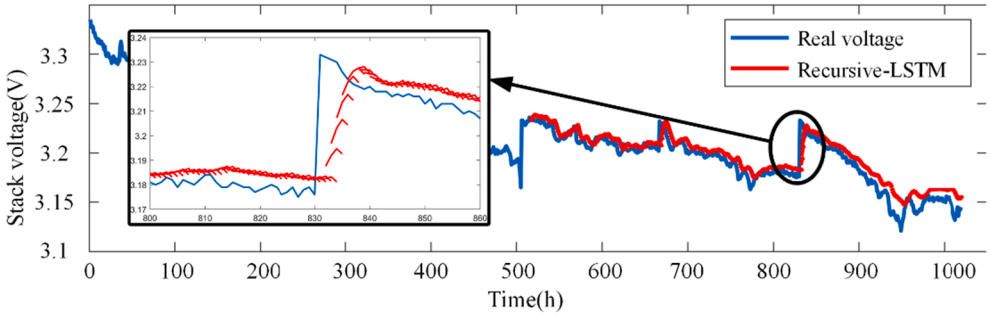
where $V_{st}$ is the stack voltage; $n$ is the number of cells; $E_0$ is the open-circuit voltage; $R$ is the total resistance; $a$ is the Tafel constant; $b$ is the concentration constant; $T$ is the stack temperature; $i$ is the stack current; $i_0$ is the exchange current, and $i_L$ is the limiting current. According to the evolution of parameters in [22], $R$ and $i_L$ are selected as time-varying parameters to characterize the aging phenomenon. Besides, the other initial parameters of the voltage degradation equation are obtained from




(a)




(b)




(c)

**Fig. 10.** The multi-step short-term forecasting results of FC2: (a) Direct strategy, (b) MIMO strategy, (c) Recursive strategy.

Pan et al. [45].

$$R(t) = R(1+\alpha(t)) \tag{17}$$

$$i_L(t) = i_L(1-\alpha(t)) \tag{18}$$

$$\alpha(t) = \int \beta(t)dt \tag{19}$$

where $\alpha$ represents the degree of aging, and $\beta$ represents the degradation rate and is nearly constant. Thus, the voltage degradation is described as a non-linear process by Eq. (20). The state equation of the system is motivated by the static response experiment. And the observation equation is modified as Eq. (23).

$$\begin{cases} X_{k+1} = AX_k + W_k \\ Y_{k+1} = g(X_k, U_k) + V_k \end{cases} \tag{20}$$

$$X_k = [\alpha_k, \beta_k]^T \tag{21}$$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \tag{22}$$

$$g(X_k, U_k) = n\left(E_0 - R(1+\alpha_k)i - aTln\left(\frac{i_k}{i_0}\right) - bTln\left(1 - \frac{i_k}{i_L(1-\alpha_k)}\right)\right) \tag{23}$$

where $X_k$ is the aging state of the system; $U_k$ is the system input; $Y_k$ is the system output; $W_k$ and $V_k$ are the process and observation noises, respectively. The procedures of the EKF algorithm are shown in Table 2.

## 4. Results and discussion

In this section, the framework of the prognostics is mainly carried out in three steps: 1) the LSTM network is applied to realize the short-term aging prognostics first and the prediction performance of five different strategies is compared under two experiment datasets of PEMFCs from different aging tests. 2) The variable-step long-term aging forecasting

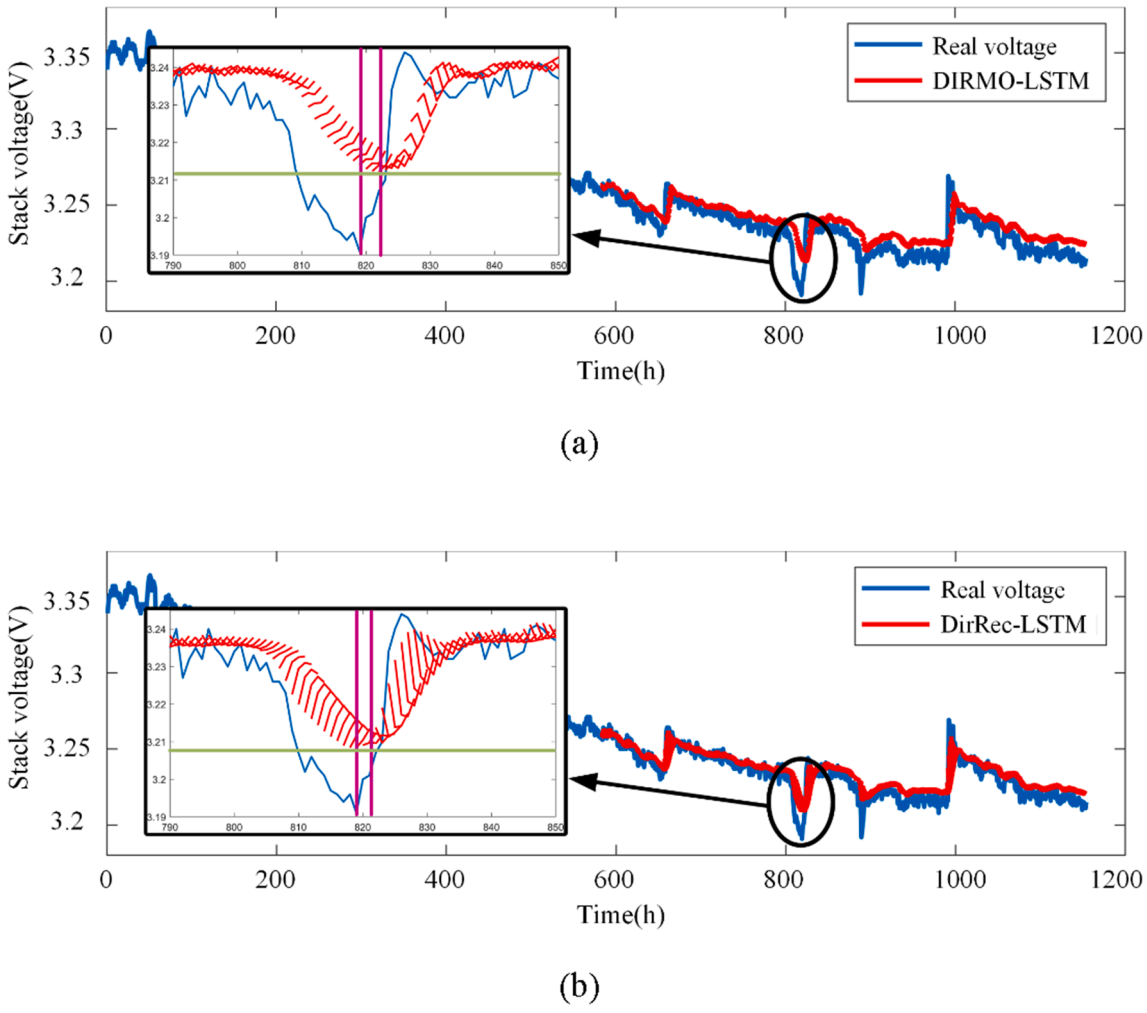



**Fig. 11.** The multi-step short-term forecasting results of FC1: (a) DIRMO strategy, (b) DirRec strategy.

method is achieved based on the MIMO strategy. 3) The EA is applied to optimize the high parameters such as the minimum batch size and the cell numbers of the hidden layers. The parameters of EA itself are shown in Table 3.

For the short-term aging prognostic, the performance of the forecasting methods over the voltage time series can be assessed by mean absolute error (MAE) and the root mean square error (RMSE) measures, defined as

$$\mathrm{MAE} = \frac{1}{H}\sum_{i=1}^{H}\left|\hat{y}_i - y_i\right| \tag{24}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{H}\sum_{i=1}^{H}\left(\hat{y}_i - y_i\right)^2} \tag{25}$$

where $H$ is the prediction horizon length in each step, $\hat{y}_i$ is the prediction and $y_i$ is the true value. And the errors can be averaged in the forecasting process to obtain a mean MAE* and a mean RMSE*, defined as

$$\mathrm{MAE}^* = \frac{1}{m}\sum_{j=1}^{m}\mathrm{MAE}_j \tag{26}$$

$$\mathrm{RMSE}^* = \frac{1}{m}\sum_{j=1}^{m}\mathrm{RMSE}_j \tag{27}$$

where $m$ is the total number of the test data. Likewise, two aging datasets of FC1 and FC2 are used to verify the effectiveness of the RUL prediction, which is defined as the difference value of the predicted EOL and the present time.

$$RUL(t) = T_{EoL}(t) - t \tag{28}$$

All the prognostics models are established in Python, and the training process is performed on a personal computer with an Intel(R) Core(TM) i7-8750H CPU (2.2 GHz clock frequency and 16 GB RAM). The RMSE is introduced to quantitatively appraise the prediction effect of the short-term aging prognostic.

Fig. 8 shows the change of training error of the MIMO-LSTM strategy during the optimization. It also can be seen from Fig. 8 that the EA approach ascertains the structural parameters of the network which converges after five iterations. The optimal structure is obtained, where the minimum batch size of each training epoch is 56, and the neuron number of three LSTM layers is 24.

### 4.1. Multi-step ahead short-term forecasting strategies comparison

To verify the prediction performance, the basic multi-step ahead strategies (Direct, MIMO, and Recursive) are firstly compared. Then, the fusion multi-step strategies are introduced to supplement the comparison. Half of the datasets are used as the training sample and the remaining sample is used as the test sample. In the training process, the historical and the prediction horizon lengths are chosen as 15 h and 4 h, respectively.

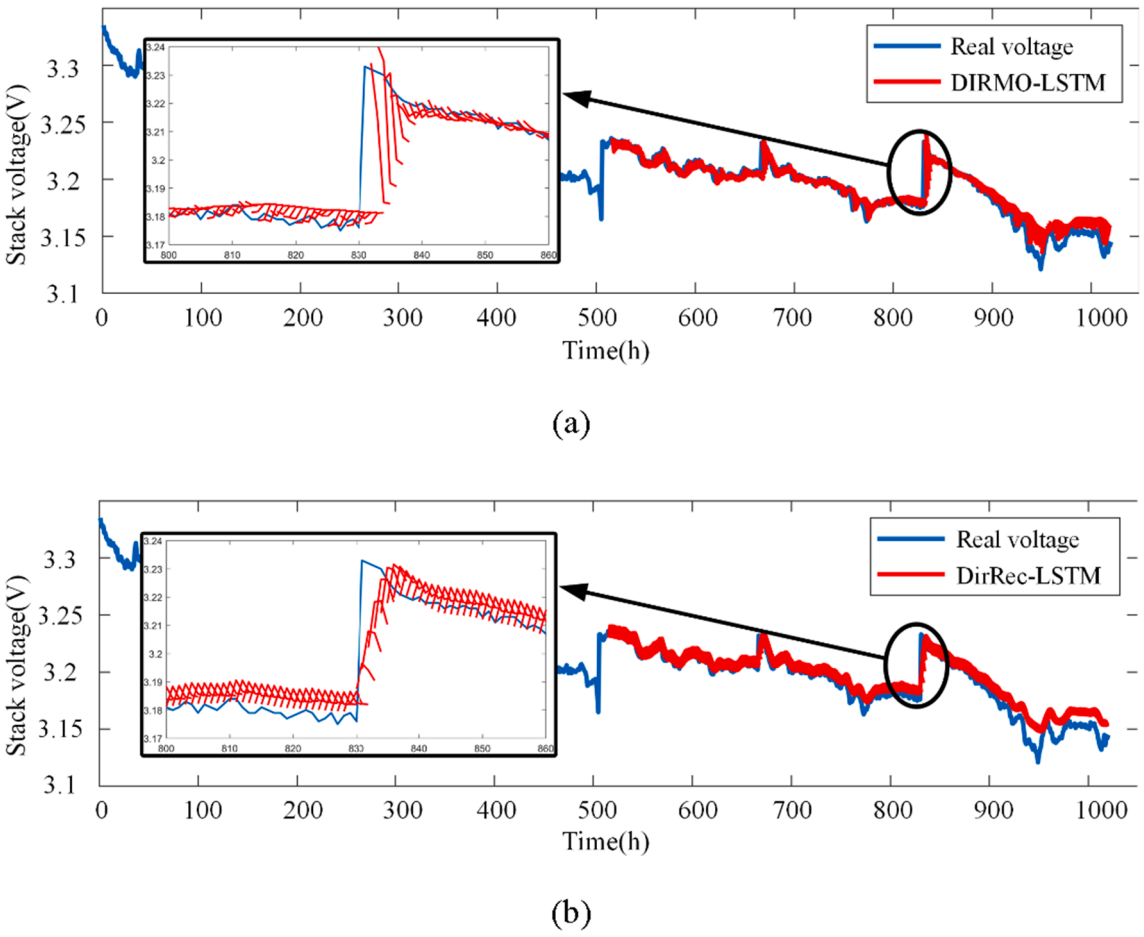



**Fig. 12.** The multi-step short-term forecasting results of FC2: (a) DIRMO strategy, (b) DirRec strategy.

**Table 4**
The comparison with different forecasting strategies of FC1.

| Strategies | Recursive | Direct | DirRec | MIMO | DIRMO |
|---|---|---|---|---|---|
| RMSE* | 0.00772 | 0.00920 | 0.00611 | 0.00532 | 0.00843 |
| MAE* | 0.00717 | 0.00831 | 0.00545 | 0.00471 | 0.00788 |
| Max RMSE | 0.0359 | 0.0366 | 0.0358 | 0.0464 | 0.0366 |
| Max MAE | 0.0350 | 0.0354 | 0.0344 | 0.0457 | 0.0359 |

**Table 5**
The comparison with different forecasting strategies of FC2.

| Strategies | Recursive | Direct | DirRec | MIMO | DIRMO |
|---|---|---|---|---|---|
| RMSE* | 0.00772 | 0.00832 | 0.00730 | 0.00538 | 0.00618 |
| MAE* | 0.00736 | 0.00763 | 0.00658 | 0.00491 | 0.00548 |
| Max RMSE | 0.0493 | 0.0486 | 0.0361 | 0.0617 | 0.0560 |
| Max MAE | 0.0492 | 0.0485 | 0.0336 | 0.0616 | 0.0543 |

(1) The basic short-term multi-step strategies

In stationary current regime, the data used for the experiment comes from an 1154 h duration test on the PEMFC stack. Half of the datasets are used for training, and the rest of the data is used for the prediction. The four-step short-term prediction results of FC1 based on the LSTM and the basic strategies are shown in Fig. 9. Under the dynamic current load, the data used for the experiment comes from an 1020 h duration test on the PEMFC stack. Half of the datasets are used for training, and the rest of the data is used for the prediction. The four-step short-term prediction results of FC2 based on the LSTM and the basic strategies are shown in Fig. 10.

The Direct-LSTM results of FC1 are shown in Fig. 9(a). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00920, 0.00831, 0.0366, and 0.0354, respectively. As shown in Fig. 9 (a), the results of the prognostic based on the Direct strategy are sharing certain biases and in larger fluctuation, especially in the middle and later periods of the prognostics. Due to the factitious intervention of the polarization test and EIS measurement, the voltage prediction results have a translation in the timeline and the *y*-direction, as shown in the detail view.

The prognostics results based on the MIMO-LSTM strategy are shown in Fig. 9(b). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00532, 0.00471, 0.0464, and 0.0457, respectively. The prognostics bias is eliminated and the MIMO strategy can get accurate prediction results in these strategies. In the detail view, the minimum predicted voltage dips below 3.201 V while the translation in the timeline remains unchanged. Furthermore, the voltage span of each prediction step is reduced even in this sharp area.

As for the Recursive-LSTM strategy, the prediction results show that the fluctuation is controlled within the satisfied region in recursion. The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00772, 0.00717, 0.0359, and 0.0350, respectively. However, due to the calculation structure, this strategy is accompanied by the prediction bias because of the accumulation of errors. In the zoomed-in view, the translation in the timeline is appreciably reduced based on the Recursive strategy. However, the minimum predicted voltage has increased to 3.213 V.

**Table 6**
Comparison of different split rate and computational complexity of the MIMO-based strategy.

| PEMFC | FC1 | | | FC2 | | |
|---|---|---|---|---|---|---|
| Split rate | 0.30 | 0.40 | 0.50 | 0.30 | 0.40 | 0.50 |
| MIMO-LSTM | | | | | | |
| RMSE* | 0.00902 | 0.00696 | 0.00532 | 0.0116 | 0.00880 | 0.00538 |
| MAE* | 0.00854 | 0.00640 | 0.00471 | 0.0112 | 0.00840 | 0.00491 |
| Computing time (s) | 13.19 | 15.45 | 20.52 | 13.61 | 16.72 | 18.86 |
| MIMO-DCNN | | | | | | |
| RMSE* | 0.00891 | 0.00659 | 0.00586 | 0.0110 | 0.00705 | 0.00652 |
| MAE* | 0.00836 | 0.00601 | 0.00527 | 0.0105 | 0.00637 | 0.00569 |
| Computing time (s) | 3.06 | 3.35 | 3.61 | 3.56 | 3.43 | 4.07 |
| MIMO-DNN | | | | | | |
| RMSE* | 0.0128 | 0.00931 | 0.00685 | 0.0126 | 0.00747 | 0.00723 |
| MAE* | 0.0123 | 0.00882 | 0.00621 | 0.0120 | 0.00681 | 0.00680 |
| Computing time (s) | 1.62 | 1.70 | 1.81 | 2.70 | 2.92 | 3.74 |

The Direct-LSTM results of FC2 are shown in Fig. 10(a). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00832, 0.00763, 0.0486, and 0.0485, respectively. As shown in Fig. 10 (a), the results of the prognostic based on the Direct strategy are sharing certain biases and in larger fluctuation, especially in the later periods of the prognostics. In the detail view, the rising fluctuation of each prediction step is unconscionable. This phenomenon is caused by the abnormal data shown in Fig. 2(b).

The prognostics results based on the MIMO-LSTM strategy are shown in Fig. 10(b). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00538, 0.00491, 0.0617, and 0.0616, respectively. The prognostics bias is eliminated and the fluctuation in the later periods is even smaller than the DIRMO strategy. The voltage recovery phenomenon is ignored by voltage prediction as a transient process. However, the real voltage is tracked by the predicted voltage within 10 steps.

As for the Recursive-LSTM strategy, the prediction results show that the fluctuation is controlled within the satisfied region in recursion. The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00772, 0.00736, 0.0493, and 0.0492, respectively. However, this strategy is accompanied by prediction bias because of the accumulation of errors. Thus, in the zoomed-in view, there remain voltage biases in the timeline and *y*-direction.

(2) The fusion short-term multi-step strategies

The four-step prognostics results of FC1 based on the DIRMO-LSTM strategy are shown in Fig. 11(a). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00843, 0.00788, 0.0366, and 0.0359, respectively. Combing with the advantage of the MIMO strategy, the fluctuation is reduced in the prediction phase, while the relentless bias is growing larger, especially after the fluctuation in 800 h. In the detail view, the minimum predicted voltage is 3.212 V.

The DirRec-LSTM results are shown in Fig. 11(b). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00611, 0.00545, 0.0358, and 0.0344, respectively. Thanks to a combination of the Direct and Recursive strategies, the bias and fluctuation of this strategy are controlled to a certain extent. However, the prediction results suffer from undesirable accuracy in the large jamming area, including 660 h, 800 h, and 1000 h. In the zoomed-in view of 800 h, the voltage span of each prediction step increases compared with other voltage gentle drop area.

The four-step prognostics results of FC2 based on the DIRMO-LSTM strategy are shown in Fig. 12(a). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00618, 0.00548, 0.0560, and 0.0543, respectively. Combing with the advantage of the MIMO strategy, the fluctuation and the bias are reduced in the prediction phase.

The DirRec-LSTM results are shown in Fig. 12(b). The RMSE*, MAE*, maximum RMSE, and maximum MAE of the test results are 0.00730, 0.00658, 0.0361, and 0.0336, respectively. Thus, the fluctuation of this strategy is controlled to a certain extent. However, the prediction results suffer from undesirable bias in the later periods of the prognostics.

The error comparison is given in Table 4. The RMSE* of the Recursive, Direct, DirRec, MIMO, and DIRMO strategies are 0.00772, 0.00920, 0.00611, 0.00532, and 0.00834. The MIMO strategy implements higher accuracy than the other four strategies in MAE* as well. However, the Recursive one has better performance in the maximum errors. The results show that the conditional independence assumption is not completely valid in the process of voltage state change. Thus, the Direct strategy has lower accuracy. The error comparison of FC2 is shown in Table 5. The MIMO strategy implements higher accuracy than the other four strategies in both RMSE* and MAE*. However, the MIMO one has unsatisfactory performance in the maximum errors, which means that the prognostics effect may be reduced in areas of large voltage variation. Similarly, the Direct strategy has the lowest accuracy as well. Furthermore, the MIMO strategy is selected to conduct the short-term and long-term prediction of FC1 and FC2.

(3) The comparison of MIMO-based strategies

As shown in Table 6, the computing time of the training process increases with the split rate. So that, these MIMO-based strategies are all available for the real practical application with the calculation burden within half a minute, while the MIMO-DCNN and the MIMO-DNN strategies suffer the lower calculation burden. Besides, the errors of RMSE and MAE show that the increase of the training datasets can improve the prediction accuracy of the MIMO-based strategy. The MIMO-LSTM strategy has the best prediction performance when the training data set is large.

### 4.2. *Variable-step long-term forecasting method result*

As for the long-term aging prognostic, the initial prediction horizon length and the prediction interval are set as 300 h and 50 h, respectively. Regarding the threshold of EoL, it is set as 3.5% of the initial voltage value ($V_{init}$ = 3.345 V) of FC1. Thus, the actual remaining useful life is 792 h in the beginning. The condition of FC2 under dynamic current load operation suffered from severer degradation than that in stationary regime. The 4.5% of the initial voltage value ($V_{init}$ = 3.330 V) is considered as the threshold of EoL. Similarly, the actual remaining useful life of FC2 is 758 h in the beginning. The variable-step long-term prognostics results of FC1 and FC2 based on MIMO-LSTM are shown in Figs. 13 and 14.

As for FC1, the prediction time values are 500 h, 550 h, 600 h, 650 h, 700 h, and 750 h, respectively. The data from 0 h to 500 h are used for training at the beginning, which is gradually extended from 0 h to 750 h

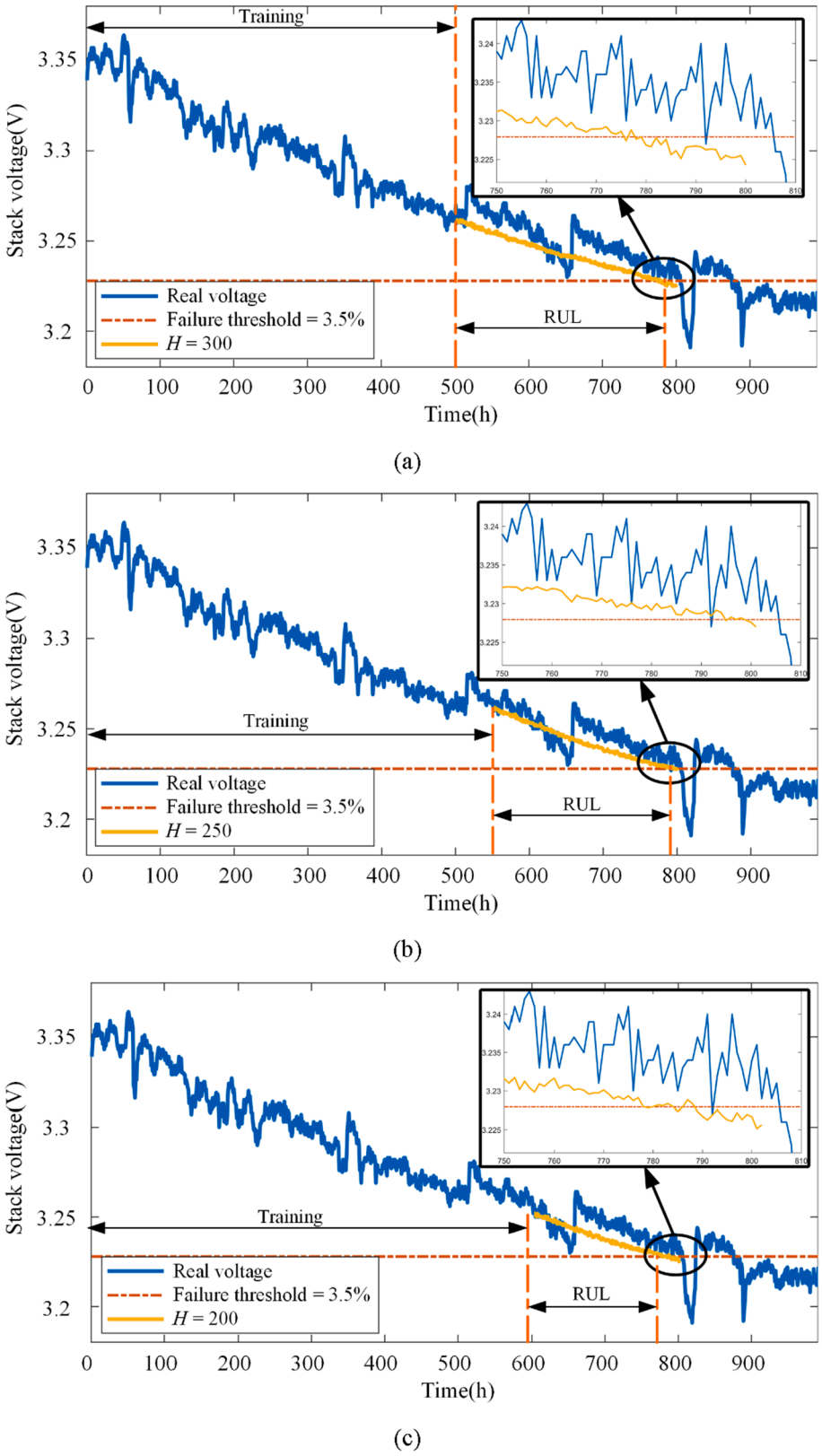
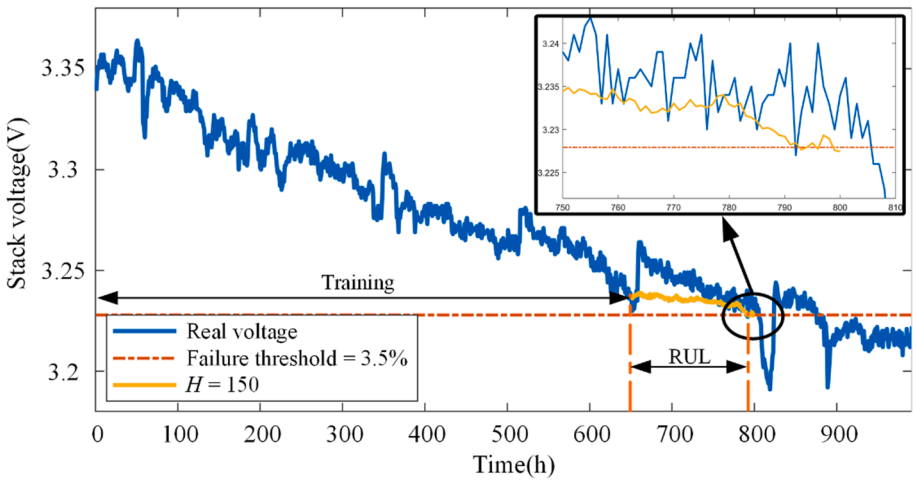
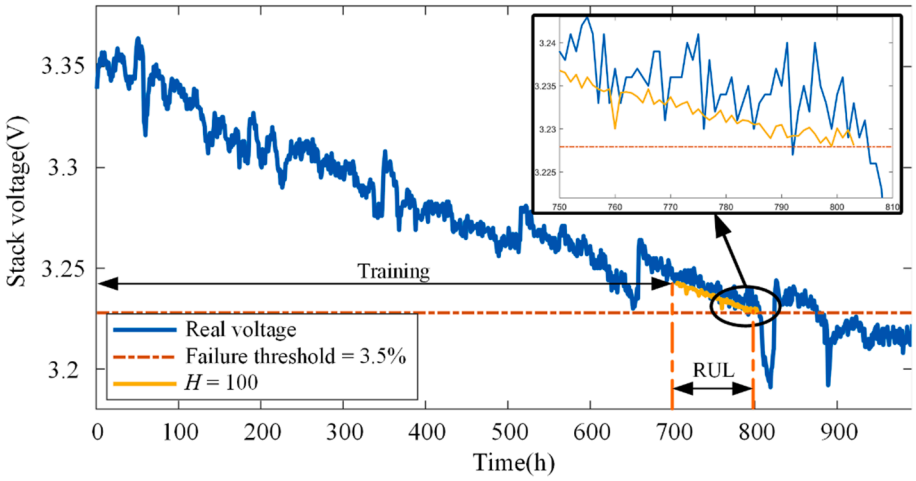



**Fig. 13.** The variable-step long-term forecasting results of FC1: (a) $H = 300$, (b) $H = 250$, (c) $H = 200$, (d) $H = 150$, (e) $H = 100$, (f) $H = 50$.

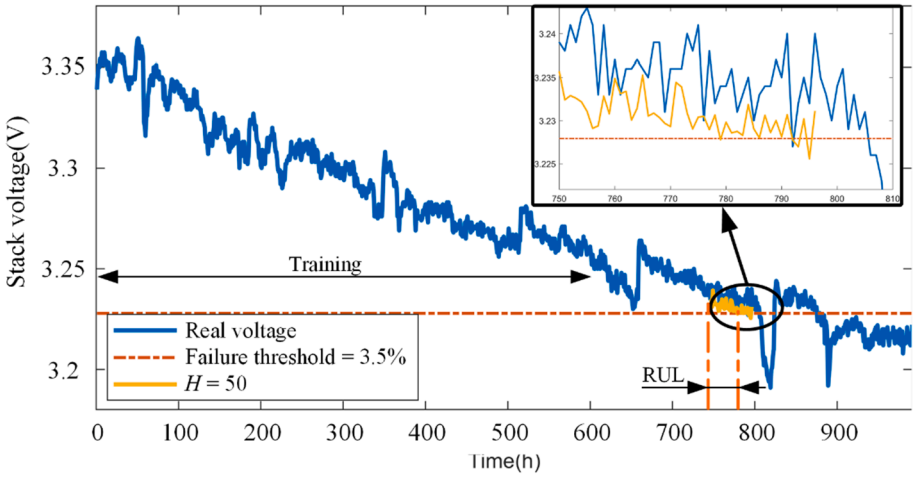
during the long-term prognostic process. As shown in Fig. 13(a)-(c), the results represent that the long-term prognostics are relatively flat when the prediction length is reduced from 300 h to 200 h. While the prediction length is reduced to 150 h, the results represent that the prediction curves have more local volatility in the details which are shown in Fig. 13(d)-(f). Due to the extended training part and the reduced prediction horizon length, the number of participating datasets is rapidly growing. Therefore, the training model can contain and




(d)




(e)

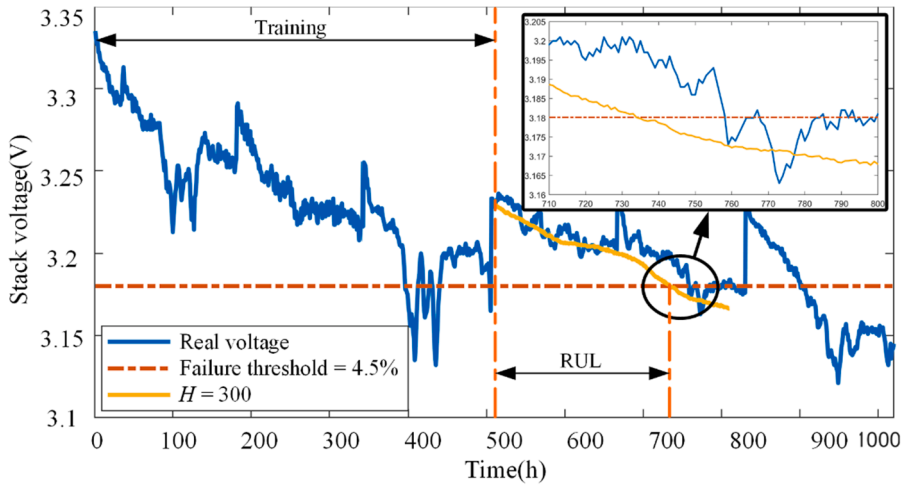



(f)

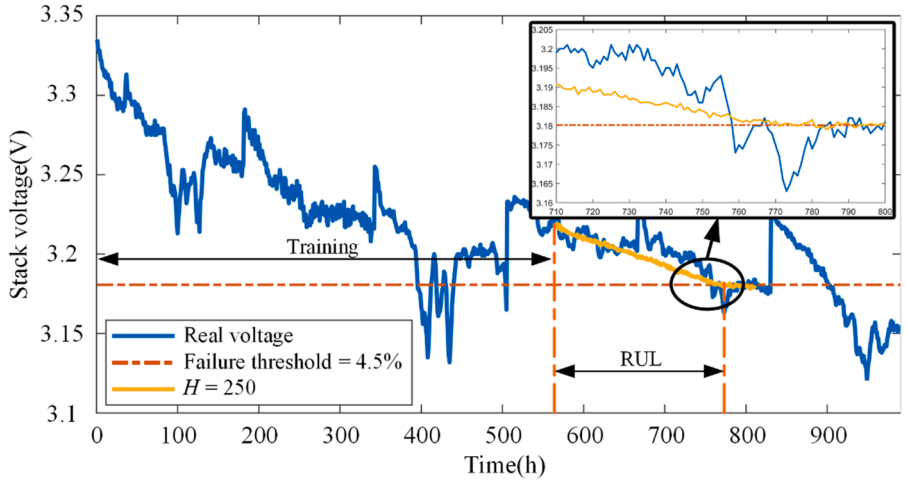
**Fig. 13.** (*continued*).

present more details of the prediction.

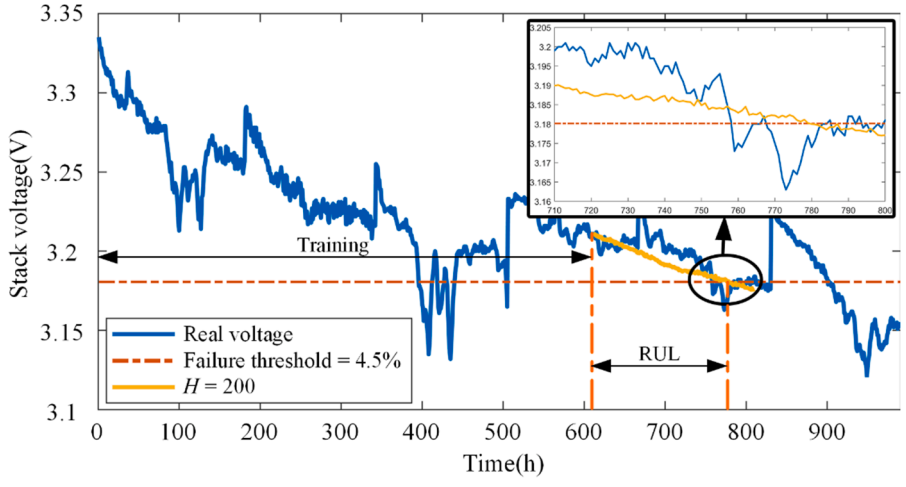
The variable-step long-term prognostics results of FC2 are shown in Fig. 14. Under the dynamic current load, the prediction time values are 510 h, 560 h, 610 h, 660 h, 710 h, and 760 h, respectively. Because the prediction process of FC2 has a more abnormal trend of change than FC1, the results also represent that the MIMO-LSTM strategy has more difficulties to mimic the dynamic current load. Especially compared with Fig. 13(a) and Fig. 14(a), the large volatility is reflected in the overall forecasting trend of FC2. The rest long-term prognostics results of FC2 are shown in Fig. 14 as well. The results present that the accuracy




(a)

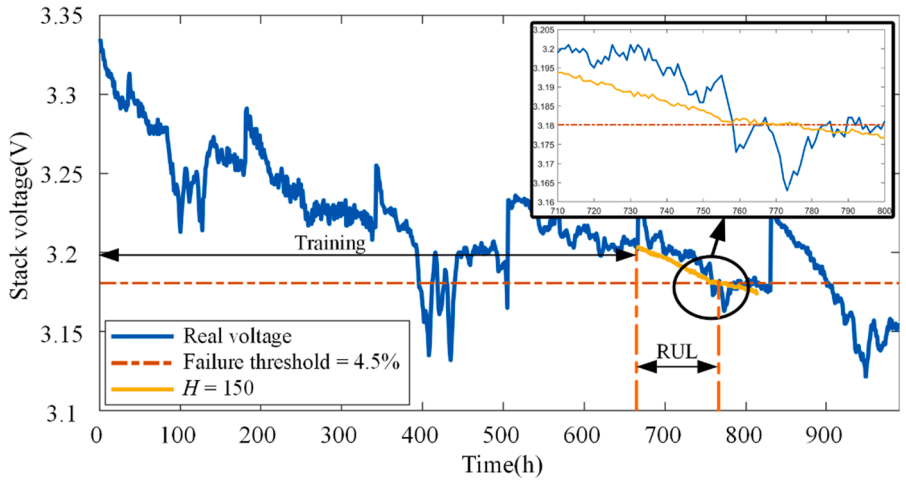



(b)

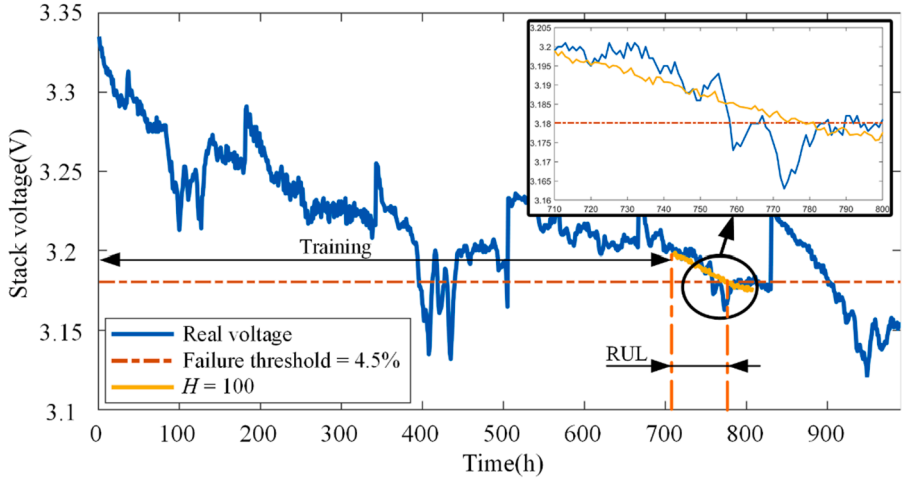



(c)

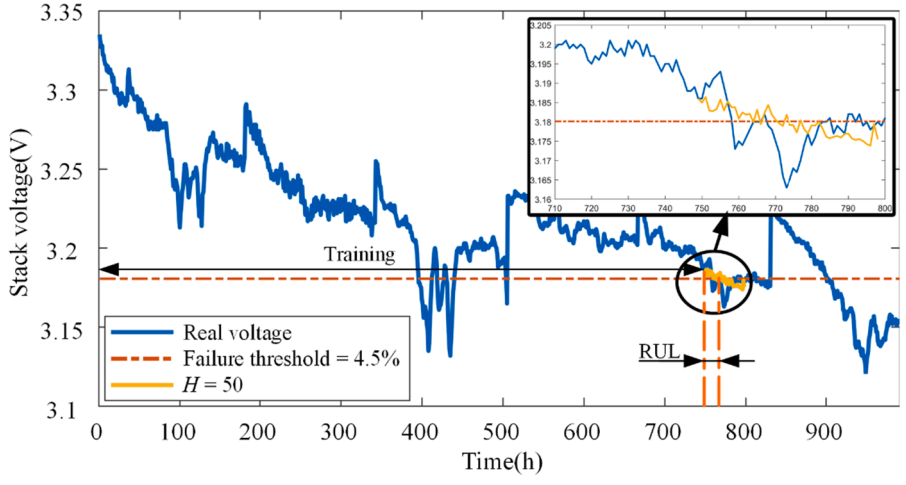
**Fig. 14.** The variable-step long-term forecasting results of FC2: (a) $H = 300$, (b) $H = 250$, (c) $H = 200$, (d) $H = 150$, (e) $H = 100$, (f) $H = 50$.




(d)




(e)




(f)

**Fig. 14.** (*continued*).

is gradually improving along with the reduction of the prediction horizon length. The reason is that the extended training sets weaken the influence of abnormal data during the training process.

The training loss and validation loss are shown in Fig. 15, where the definition is shown in Eq. (24). MIMO-LSTM strategy has a fast convergence rate, which has begun to converge at the 8th epoch. In each epoch, the validation loss is greater than the training one, indicating no overfitting phenomena. The results show that MIMO-LSTM has a fast constringency speed and good generalization performance.

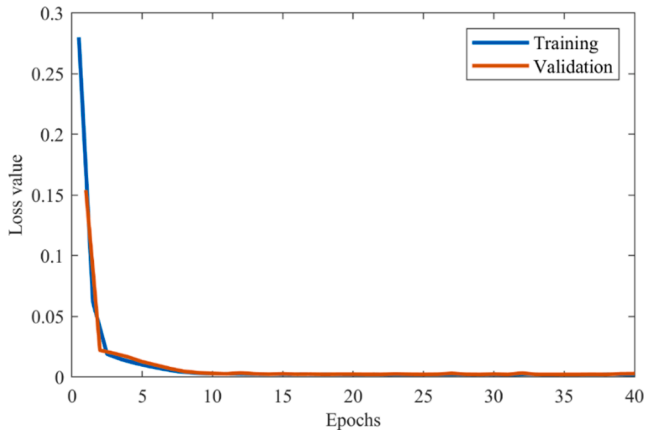



**Fig.15.** The training process of the proposed mothed.

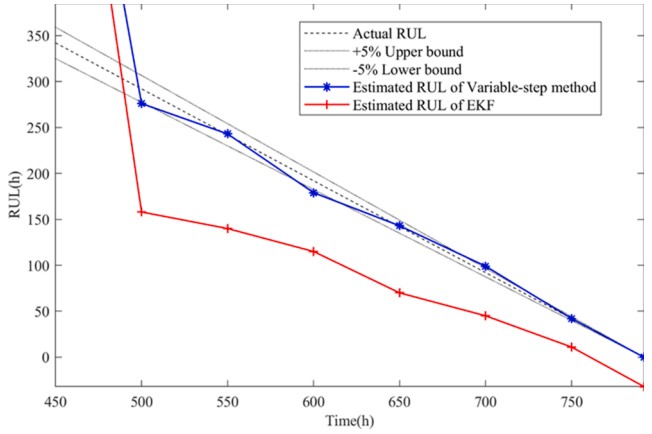



**Fig. 16.** The RUL prediction of FC1 based on the variable-step long-term method and EKF method.

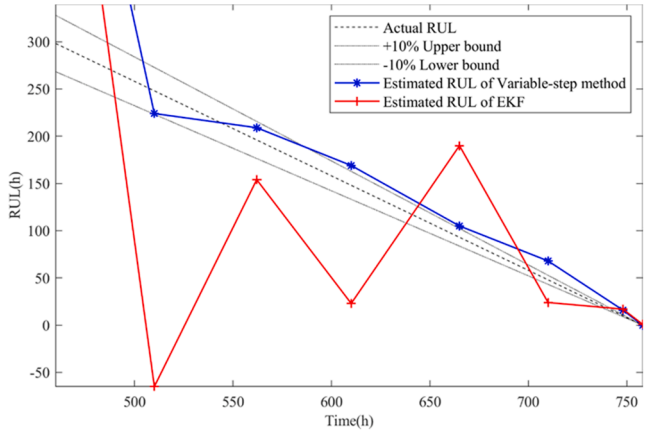



**Fig. 17.** The RUL prediction of FC2 based on the variable-step long-term method and EKF method.

### *4.3. RUL prediction comparison*

The RUL time here is considered as the time between the prediction time and the failure time. And the RUL prediction result of the proposed method is compared with the EKF method. Meanwhile, several points are used to construct these evolutions. The RUL prediction results of FC1 and FC2 based on the variable-step long-term method and EKF method are shown in Figs. 16 and 17.

To obtain satisfactory results, the estimated RUL should be within the error bounds of the real RUL. In Fig. 16, the estimated RUL of the variable-step MIMO-LSTM method is within the bounds ±5% of the real RUL, while the estimated RUL of EKF is out of the bounds until the last estimation. As shown in Fig. 17, the estimated RUL of FC2 based on the variable-step MIMO-LSTM method is within the bounds of the real RUL as well. However, the RUL results remain more errors compared with the FC1. Especially for the RUL estimation of FC2 based on EKF, there remain large fluctuations in the forecasting points which are out of the ±10% bounds. The reason is that the state equation of the EKF is motivated by the linear evolution of the aging parameters, while the system aging parameters shall contain more nonlinear factors under the dynamic current load. Thus, the RUL prediction performance of FC2 based on the EKF is not satisfied.

## 5. Conclusion

This paper has proposed a data-based prognostics framework to achieve the multi-step ahead short-term forecasting and the variable-step long-term prediction at the same time. The proposed framework can provide a concise forecasting of PEMFC output performance without the future information as prior knowledge. First, five multi-step strategies based on an LSTM structure are developed and compared by two data sets from the stationary and dynamic aging process. The comparison results figure out that the MIMO strategy is the most feasible strategy for the multi-step short-term prognostics of PEMFC. For the FC1 and FC2 in this study, the best short-term prognostics performance yielded by the MIMO-LSTM in terms of RMSE* is 0.00532 and 0.00538, respectively, and in terms of MAE* is 0.00471 and 0.00491, respectively. Then, a hyper-parameters adjustment method is proposed to improve the training parameters of network structure. A variable-step long-term prognostics method is subsequently investigated based on the MIMO-LSTM. Moreover, the model-based EKF prognostics are compared with the proposed long-term prognostics method. The results indicate that the proposed RUL estimation accuracy can be improved to 95% and 90% confidence intervals for FC1 and FC2, respectively. Therefore, the proposed short- and long-term prognostics framework has a significant performance in the short-term, long-term, and RUL prediction. However, faced with a more complex operating condition and aging process, a data-driven prediction method is difficult to be directly transferred into the specific application. Furthermore, future work can be extended to combine with the model-based method to expand its adaptability, in particular to solve the developed data-based method drawback of dynamic performance forecasting relatively low accuracy.

## CRediT authorship contribution statement

**Zhendong Zhang:** Methodology, Software, Formal analysis, Writing – original draft. **Ya-Xiong Wang:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Hongwen He:** Validation, Investigation, Supervision. **Fengchun Sun:** Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This work was supported by the China Postdoctoral Science Foundation (Grant No. 2019M650505).

## References


[1] Wang X, Chen J, Quan S, Wang YX, He H. Hierarchical model predictive control via deep learning vehicle speed predictions for oxygen stoichiometry regulation of fuel cells. Appl Energy 2020;276:115460.
[2] Jouin M, Gouriveau R, Hissel D, Péra M-C, Zerhouni N. Prognostics and health management of PEMFC – state of the art and remaining challenges. Int J Hydrogen Energy 2013;38(35):15307–17.
[3] Liu H, Chen J, Hissel D, Lu J, Hou M, Shao Z. Prognostics methods and degradation indexes of proton exchange membrane fuel cells: a review. Renew Sustain Energy Rev 2020;123:109721. https://doi.org/10.1016/j.rser.2020.109721.
[4] Marra D, Sorrentino M, Pianese C, Iwanschitz B. A neural network estimator of solid oxide fuel cell performance for on-field diagnostics and prognostics applications. J Power Sources 2013;241:320–9.
[5] Silva RE, Gouriveau R, Jemeï S, Hissel D, Boulon L, Agbossou K, et al. Proton exchange membrane fuel cell degradation prediction based on adaptive Neuro-Fuzzy inference systems. Int J Hydrogen Energy 2014;39(21):11128–44.
[6] Morando S, Jemei S, Hissel D, Gouriveau R, Zerhouni N. Proton exchange membrane fuel cell ageing forecasting algorithm based on Echo State Network. Int J Hydrogen Energy 2017;42(2):1472–80.
[7] Hua Z, Zheng Z, Péra M, Gao F. Remaining useful life prediction of PEMFC systems based on the multi-input echo state network. Appl Energy 2020;265:114791.
[8] Wu Y, Breaz E, Gao F, Paire D, Miraoui A. Nonlinear performance degradation prediction of proton exchange membrane fuel cells using relevance vector machine. IEEE Trans Energy Convers 2016;31(4):1570–82.
[9] Wu Y, Breaz E, Gao F, Miraoui A. A modified relevance vector machine for PEM fuel-cell stack aging prediction. IEEE Trans Ind Appl 2016;52(3):2573–81.
[10] Kim J, Lee I, Tak Y, Cho BH. State-of-health diagnosis based on hamming neural network using output voltage pattern recognition for a PEM fuel cell. Int J Hydrogen Energy 2012;37(5):4280–9.
[11] Zhou D, Al-Durra A, Zhang Ke, Ravey A, Gao F. Online remaining useful lifetime prediction of proton exchange membrane fuel cells using a novel robust methodology. J Power Sources 2018;399:314–28.
[12] Liu H, Chen J, Hou M, Shao Z, Su H. Data-based short-term prognostics for proton exchange membrane fuel cells. Int J Hydrogen Energy 2017;42(32):20791–808.
[13] Liu H, Chen J, Hissel D, Su H. Short-term prognostics of PEM fuel cells: a comparative and improvement study. IEEE Trans Ind Electron 2019;66(8):6077–86.
[14] Li Z, Zheng Z, Outbib R. Adaptive prognostic of fuel cells by implementing ensemble echo state networks in time-varying model space. IEEE Trans Ind Electron 2020;67(1):379–89.
[15] Chen K, Laghrouche S, Djerdir A. Health state prognostic of fuel cell based on wavelet neural network and cuckoo search algorithm. ISA Trans 2021;113:175–84.
[16] Xie Y, Zou J, Li Z, Gao F, Peng C. A novel deep belief network and extreme learning machine based performance degradation prediction method for proton exchange membrane fuel cell. IEEE Access 2020;8:176661–75.
[17] Meraghni S, Terrissa LS, Yue M, Ma J, Jemei S, Zerhouni N. A data-driven digital-twin prognostics method for proton exchange membrane fuel cell remaining useful life prediction. Int J Hydrogen Energy 2021;46(2):2555–64.
[18] Zuo J, Lv H, Zhou D, Xue Q, Jin L, Zhou W, et al. Deep learning based prognostic framework towards proton exchange membrane fuel cell for automotive application. Appl Energy 2021;281:115937.
[19] Ma J, Liu X, Zou X, Yue M, Shang P, Kang L, et al. Degradation prognosis for proton exchange membrane fuel cell based on hybrid transfer learning and intercell differences. ISA Trans 2021;113:149–65.
[20] He K, Mao L, Yu J, Huang W, He Q, Jackson L. Long-term performance prediction of PEMFC based on LASSO-ESN. IEEE Trans Instrum Meas 2021;70:1–11.
[21] Jouin M, Gouriveau R, Hissel D, Péra M-C, Zerhouni N. Prognostics of PEM fuel cell in a particle filtering framework. Int J Hydrogen Energy 2014;39(1):481–94.
[22] Bressel M, Hilairet M, Hissel D, Ould BB. Extended kalman filter for prognostic of proton exchange membrane fuel cell. Appl Energy 2016;164:220–7.
[23] Zhou D, Wu Y, Gao F, Breaz E, Ravey A, Miraoui A. Degradation prediction of PEM fuel cell stack based on multiphysical aging model with particle filter approach. IEEE Trans Ind Appl 2017;53(4):4041–52.
[24] Jouin M, Gouriveau R, Hissel D, Pera M-C, Zerhouni N. Joint particle filters prognostics for proton exchange membrane fuel cell power prediction at constant current solicitation. IEEE Trans Reliab 2016;65(1):336–49.
[25] Zhu Li, Chen J. Prognostics of PEM fuel cells based on Gaussian process state space models. Energy 2018;149:63–73.
[26] Bressel M, Hilairet M, Hissel D, Bouamama BO. Remaining useful life prediction and uncertainty quantification of proton exchange membrane fuel cell under variable load. IEEE Trans Ind Electron 2016;63(4):2569–77.
[27] Jha MS, Bressel M, Ould-Bouamama B, Dauphin-Tanguy G. Particle filter based hybrid prognostics of proton exchange membrane fuel cell in bond graph framework. Comput Chem Eng 2016;95:216–30.
[28] Robin C, Gerard M, Franco AA, Schott P. Multi-scale coupling between two dynamical models for PEMFC aging prediction. Int J Hydrogen Energy 2013;38(11):4675–88.
[29] Chen J, Zhou D, Lyu C, Lu C. A novel health indicator for PEMFC state of health estimation and remaining useful life prediction. Int J Hydrogen Energy 2017;42(31):20230–8.
[30] Zhang D, Baraldi P, Cadet C, Yousfi-Steiner N, Bérenguer C, Zio E. An ensemble of models for integrating dependent sources of information for the prognosis of the remaining useful life of Proton Exchange Membrane Fuel Cells. Mech Syst Sig Process 2019;124:479–501.
[31] Robin C, Gérard M, Quinaud M, d'Arbigny J, Bultel Y. Proton exchange membrane fuel cell model for aging predictions: Simulated equivalent active surface area loss and comparisons with durability tests. J Power Sources 2016;326:417–27.
[32] Ou M, Zhang R, Shao Z, Li B, Yang D, Ming P, et al. A novel approach based on semi-empirical model for degradation prediction of fuel cells. J Power Sources 2021;488:229435. https://doi.org/10.1016/j.jpowsour.2020.229435.
[33] Chen K, Laghrouche S, Djerdir A. Fuel cell health prognosis using Unscented Kalman Filter: Postal fuel cell electric vehicles case study. Int J Hydrogen Energy 2019;44(3):1930–9.
[34] Wu X, Ye Q, Wang J. A hybrid prognostic model applied to SOFC prognostics. Int J Hydrogen Energy 2017;42(39):25008–20.
[35] Cheng Y, Zerhouni N, Lu C. A hybrid remaining useful life prognostic method for proton exchange membrane fuel cell. Int J Hydrogen Energy 2018;43(27):12314–27.
[36] Chen K, Laghrouche S, Djerdir A. Degradation model of proton exchange membrane fuel cell based on a novel hybrid method. Appl Energy 2019;252:113439.
[37] Zhou D, Gao F, Breaz E, Ravey A, Miraoui A. Degradation prediction of PEM fuel cell using a moving window based hybrid prognostic approach. Energy 2017;138:1175–86.
[38] Xie R, Ma R, Pu S, Xu L, Zhao D, Huangfu Y. Prognostic for fuel cell based on particle filter and recurrent neural network fusion structure. Energy AI 2020;2:100017. https://doi.org/10.1016/j.egyai.2020.100017.
[39] Zuo B, Cheng J, Zhang Z. Degradation prediction model for proton exchange membrane fuel cells based on long short-term memory neural network and Savitzky-Golay filter. Int J Hydrogen Energy 2021;46(29):15928–37.
[40] Liu J, Li Qi, Chen W, Yan Yu, Qiu Y, Cao T. Remaining useful life prediction of PEMFC based on long short-term memory recurrent neural networks. Int J Hydrogen Energy 2019;44(11):5470–80.
[41] He K, Zhang C, He Q, Wu Q, Jackson L, Mao L. Effectiveness of PEMFC historical state and operating mode in PEMFC prognosis. Int J Hydrogen Energy 2020;45(56):32355–66.
[42] Hua Z, Zheng Z, Pahon E, Péra M, Gao F. Remaining useful life prediction of PEMFC systems under dynamic operating conditions. Energy Convers Manage 2021;231:113825.
[43] FCLAB Research. IEEE PHM 2014 data challenge. 2014. http://eng.fclab.fr/ieee-phm-2014-data-challenge/.
[44] Taieb SB, Bontempi G, Atiya AF, Sorjamaa A. A review and comparison of strategies for multi-step ahead time series forecasting based on the NN5 forecasting competition. Expert Syst Appl 2012;39(8):7067–83.
[45] Pan R, Yang D, Wang Y, Chen Z. Performance degradation prediction of proton exchange membrane fuel cell using a hybrid prognostic approach. Int J Hydrogen Energy 2020;45(55):30994–1008.